

Unit 1 - Math Fundamentals

Sets

MSDS 520

Meharry Medical College

Outline

- 1 Why Sets?
- 2 Sets and Membership
- 3 Subsets
- 4 Set Operations
- 5 Venn Diagrams and Counting
- 6 Sets of Numbers
- 7 Intervals
- 8 Quantifiers
- 9 Ordered Pairs and Tuples
- 10 Review

Why Sets?

A Problem to Motivate Us

A question about a patient population

*“How many patients have diabetes **or** hypertension, but **not both**?”*

To answer this we need language for:

- a **collection** of patients (“the diabetics”),
- **combining** collections (“diabetic or hypertensive”),
- **removing** one collection from another (“but not both”),
- and **counting** what is left.

That language is **set theory**. It is the same logic as Unit 1-1 — AND, OR, NOT — but now applied to collections of things rather than to statements.

Where This Shows Up in Data Science

Sets are everywhere once you look

- **Cohorts and subgroups:** every “group of patients” is a set.
- **Joining data:** merging two tables is an intersection of their keys.
- **Domains of functions:** the set of inputs a function accepts.
- **Reference ranges:** “normal” lab values are an **interval** — a set of numbers.
- **Probability:** an event is a set of outcomes (Unit 4).

What we will build

- ① **Sets**, membership, and **subsets**.
- ② The **operations**: union, intersection, complement.
- ③ **Venn diagrams** and counting.
- ④ The standard **sets of numbers** and **intervals**.
- ⑤ **Quantifiers** — how to say “for all” and “there exists”.

Sets and Membership

What Is a Set?

Set

A **set** is a collection of objects, called its **elements** (or **members**).

- Sets are usually named with capital letters: A, B, C .
- Elements are listed inside curly braces:

$$A = \{1, 2, 3\}$$

- **Order does not matter:** $\{1, 2, 3\}$ and $\{3, 1, 2\}$ are the same set.
- **Repeats do not matter:** $\{1, 1, 2\}$ is just $\{1, 2\}$.

A set is defined entirely by *what is in it* — nothing else.

Membership

To say that an object is in a set, we use the symbol $\in$:

$$1 \in A \quad \text{"1 is an element of } A\text{"}$$

And to say it is not, we strike it through:

$$5 \notin A \quad \text{"5 is not an element of } A\text{"}$$

Connection to Unit 1-1

" $x \in A$ " is a **statement** — it is true or false, and so it can be combined with $\neg$, $\wedge$, $\vee$ like any other. Notice that its truth depends on which x you pick; we give that idea a name and a notation later in this unit. Everything we do with sets will mirror something we did with logic.

Some Special Sets

Singletons

A set with exactly one element, e.g. $\{1\}$, is called a **singleton**.

Note that 1 and $\{1\}$ are different things: a number, and a box containing that number.

The empty set

The set with **no** elements is called the **empty set**:

$$\emptyset = \{ \}$$

A universe

Often we fix a **universal set** U containing everything under discussion — e.g. all patients in the study. “Everything else” always means “everything else in U .”

Set-Builder Notation

Listing elements only works for small sets. For large or infinite ones, we describe the **rule** that members satisfy:

$$A = \{ x \mid x > 0 \} \quad \text{“the set of all positive numbers”}$$

Read the vertical bar (pipe) as “**such that**”:

“the set of all x *such that* x is greater than 0”

More examples

- $\{ x \mid x \text{ is a day of the week} \}$
- $\{ x \mid x \text{ is a patient and } x \text{ is diabetic} \}$
- $\{ x \in U \mid x \text{ is diabetic} \}$ (membership can be stated up front)

The rule after the bar is a **statement about** x — logic again.

Exercise: Translating Set-Builder Notation

Part 1 — translate set-builder notation into plain English

- ① $\{x \mid x < 0\}$
- ② $\{x \in U \mid x \text{ is pregnant}\}$ ($U = \text{all patients in the study}$)
- ③ $\{x \mid x \text{ is a month with exactly 30 days}\}$ — then list it.

Part 2 — now go the other way, English into set-builder notation

- ① The set of all patients who are diabetic **and** over 65.
- ② The set of all letters that are vowels.
- ③ The set of all numbers between 10 and 20.

Exercise: Solution

Part 1 — into English

- ① “the set of all x *such that* x is less than 0” — the negative numbers.
- ② “the set of all patients in the study who are pregnant.”
- ③ “the set of all months with exactly 30 days”
= {April, June, September, November}.

Part 2 — into set-builder

- ① $\{x \in U \mid x \text{ is diabetic} \wedge x \text{ is over } 65\}$ (the AND from Unit 1-1)
- ② $\{x \mid x \text{ is a vowel}\} = \{a, e, i, o, u\}$
- ③ $\{x \mid 10 < x < 20\}$

That last one is slippery: “between” never says whether 10 and 20 themselves count. **Intervals** will pin this down.

Cardinality

Cardinality

The **cardinality** of a set A , written $|A|$, is the number of elements it contains.

Examples

- $A = \{1, 2, 3\} \Rightarrow |A| = 3$
- $|\{7\}| = 1$
- $|\emptyset| = 0$

Every “how many patients . . .” question is a cardinality question. We will come back to this when we count with Venn diagrams.

Exercise

Try it

Let

$$A = \{2, 4, 6, 8\}$$

$$B = \{x \mid x \text{ is a letter in the word "level"}\}$$

- ① Write B by listing its elements.
- ② Find $|A|$ and $|B|$.
- ③ True or false: $4 \in A$, $\{4\} \in A$, $5 \notin A$.

Exercise: Solution

1. Listing B

The letters of “level” are l, e, v, e, l — but **repeats do not matter**, so $B = \{l, e, v\}$.

2. Cardinality

$|A| = 4$ and $|B| = 3$ — three *distinct* letters, not five.

3. True or false

- $4 \in A$ — **true**, 4 is one of the listed elements.
- $\{4\} \in A$ — **false**. The elements of A are numbers; $\{4\}$ is a *set*, and no set appears in A .
- $5 \notin A$ — **true**, 5 is not listed.

Subsets

Subsets

Subset

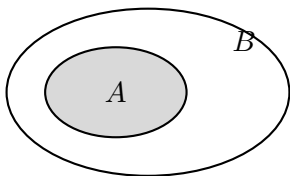
A is a **subset** of B — written $A \subseteq B$ — if **every** element of A is also an element of B .

Written with the implication from Unit 1-1 — for every x :

$$x \in A \rightarrow x \in B$$

Example

$\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2, 3\} \subseteq \{1, 2, 3\}$ — a set is a subset of itself.



Proper Subsets, and the Empty Set

Proper subset

A is a **proper** subset of B — written $A \subset B$ — if $A \subseteq B$ **and** $A \neq B$. That is, B has at least one element that A does not.

$A \subseteq B$	contained in, possibly equal
$A \subset B$	contained in, definitely smaller

Convention: $\emptyset \subseteq B$ for every set B

The empty set is a subset of **everything**.

Why? To fail, $\emptyset$ would need an element that is not in B — and it has no elements at all to offer. The condition is **vacuously true** (Unit 1-1).

Set Equality

Two sets are equal when each contains the other

$$A = B \quad \text{means} \quad A \subseteq B \quad \text{and} \quad B \subseteq A$$

This is the set-theory version of the bi-implication $\leftrightarrow$ from Unit 1-1:
 $x \in A$ if and only if $x \in B$.

This is also a proof strategy

To prove two sets are the same, prove **both** containments:

- ① Take an arbitrary $x \in A$ and show $x \in B$.
- ② Take an arbitrary $x \in B$ and show $x \in A$.

Exercise

Try it

Let $A = \{1, 2, 3\}$.

- ① List **all** subsets of A . How many are there?
- ② Which of those are **proper** subsets?
- ③ True or false: $\emptyset \subseteq A$, $\emptyset \in A$, $A \subseteq A$, $A \subset A$.

Exercise: Solution

1. All subsets — eight of them

$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

A set with n elements has 2^n subsets — here $2^3 = 8$.

2. Proper subsets

Every one of them except $\{1, 2, 3\}$ itself — **seven**.

3. True or false

- $\emptyset \subseteq A$ — **true**. The empty set is a subset of everything.
- $\emptyset \in A$ — **false**. The elements of A are 1, 2, 3; $\emptyset$ is not one of them.
- $A \subseteq A$ — **true**. $A \subset A$ — **false**: proper demands $A \neq A$.

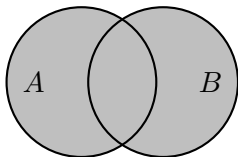
Set Operations

Union

Union

The **union** of A and B , written $A \cup B$, is the set of everything in A , in B , or in both:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$



$$\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}$$

Note

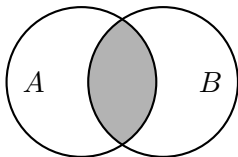
Union is the set version of the logical **OR** — and it is the *inclusive* OR: elements in both sets are included, listed once.

Intersection

Intersection

The **intersection** of A and B , written $A \cap B$, is the set of everything in **both**:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



$$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}$$

Note

Intersection is the set version of the logical **AND**.

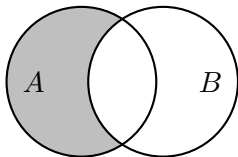
If $A \cap B = \emptyset$, the sets share nothing and are called **disjoint**.

Difference (Relative Complement)

Set difference

The **difference** $A \setminus B$ is everything in A that is **not** in B :

$$A \setminus B = \{x \in A \mid x \notin B\}$$



Note

Difference is the set version of **AND NOT**. **Order matters:**

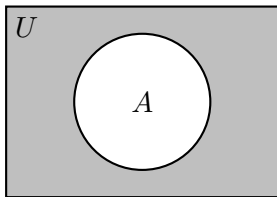
$$\{1, 2, 3\} \setminus \{2, 3, 4\} = \{1\}$$

$$\{2, 3, 4\} \setminus \{1, 2, 3\} = \{4\}$$

Complement

When a universal set U is fixed, the **complement** of A is everything in U outside of A :

$$\bar{A} = U \setminus A = \{x \in U \mid x \notin A\}$$



Note

Complement is the set version of **NOT**.

It only makes sense once U is agreed upon — “not diabetic” means nothing until we say “among whom.”

Some texts write A' instead of $\bar{A}$.

Worked Examples

Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$.

- $A \cup B = \{1, 2, 3, 4\}$

- $A \cap B = \{2, 3\}$

- $A \setminus B = \{1\}$

- $B \setminus A = \{4\}$

Exercise

Try it

With $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$, find:

$$A \cap C, \quad (A \cup B) \cap C, \quad A \setminus (B \cap C)$$

Exercise: Solution

$A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$. Work from the inside of the parentheses outward.

$$A \cap C$$

Elements in both A and C . Only 3 qualifies: $A \cap C = \{3\}$

$$(A \cup B) \cap C$$

First the inside: $A \cup B = \{1, 2, 3, 4\}$.

Then intersect with $C = \{3, 4, 5\}$: $(A \cup B) \cap C = \{3, 4\}$

$$A \setminus (B \cap C)$$

First the inside: $B \cap C = \{3, 4\}$.

Then remove those from $A = \{1, 2, 3\}$ — the 4 was not there to remove:

$$A \setminus (B \cap C) = \{1, 2\}$$

Properties of Union and Intersection

These behave much like $+$ and $\times$ in ordinary algebra.

Commutative — order does not matter

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative — grouping does not matter

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$\cup$ and $\cap$ are **binary** — each joins two sets at a time. Associativity is what lets us drop the parentheses and write $A \cup B \cup C$, exactly as for $1 + 2 + 3$.

Distributive — each distributes over the other

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

The first of these has no arithmetic analogue — $+$ does not distribute over $\times$.

De Morgan's Laws for Sets

The laws from Unit 1-1 carry over word for word, with $\cup$ for OR and $\cap$ for AND.

De Morgan's Laws (sets)

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Why they are the same laws

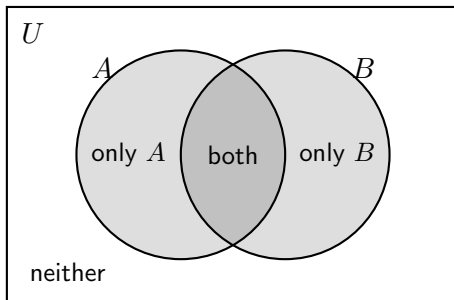
$$\begin{aligned} x \in \overline{A \cap B} &\Leftrightarrow \neg(x \in A \wedge x \in B) \\ &\Leftrightarrow \neg(x \in A) \vee \neg(x \in B) && \text{(logical De Morgan)} \\ &\Leftrightarrow x \in \bar{A} \cup \bar{B} \end{aligned}$$

Set theory is logic with the statements collected into piles.

Venn Diagrams and Counting

Reading a Two-Set Venn Diagram

Two overlapping circles cut the universe into **four** regions — and every patient sits in exactly one of them.



only A	$A \setminus B$
both	$A \cap B$
only B	$B \setminus A$
neither	$\overline{A \cup B} = \bar{A} \cap \bar{B}$

Inclusion–Exclusion

How many elements are in $A \cup B$? Adding $|A|$ and $|B|$ counts the overlap **twice**, so we subtract it back off once:

Inclusion–exclusion (two sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example

In a clinic of 100 patients, let D be the set of diabetics and H the set of hypertensives. Suppose $|D| = 40$, $|H| = 55$, and $|D \cap H| = 25$.

$$|D \cup H| = 40 + 55 - 25 = 70$$

So 70 patients have at least one condition — and $100 - 70 = 30$ have neither.

Notice that $40 + 55 = 95 \neq 70$. Forgetting to subtract the overlap is the classic

Answering the Opening Question

The question

*“How many patients have diabetes **or** hypertension, but **not both**?”*

In symbols, that is the region $(D \cup H) \setminus (D \cap H)$ — the two crescents, but not the middle:

$$|(D \cup H) \setminus (D \cap H)| = |D \cup H| - |D \cap H|$$

With the numbers from the previous slide:

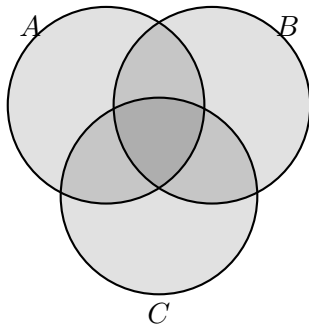
$$70 - 25 = 45$$

Recognize the shape

“Or, but not both” is the **exclusive OR** from Unit 1-1. The same idea, now counted.

Three-Set Venn Diagrams

Three circles cut the universe into **eight** regions (2^3 — one for each in/out combination, exactly like a truth table).



Strategy: fill in from the middle outward

Start with $|A \cap B \cap C|$, then the two-way overlaps (subtracting the center), then the single regions. Each region gets counted exactly once.

Exercise

Try it

A clinic surveys 120 patients. Let E be the set who exercise regularly, and D the set who follow a special diet:

$$|E| = 70, \quad |D| = 45, \quad |E \cap D| = 25$$

- ① How many do **at least one** of the two?
- ② How many do **neither**?
- ③ How many exercise but do **not** diet?

Exercise: Solution

1. At least one — inclusion–exclusion

$$|E \cup D| = |E| + |D| - |E \cap D| = 70 + 45 - 25 = 90$$

2. Neither — everything outside that union

$$|\overline{E \cup D}| = 120 - 90 = 30$$

3. Exercise but not diet — a difference

$$|E \setminus D| = |E| - |E \cap D| = 70 - 25 = 45$$

The 25 who do both are already counted inside E , so subtract them off.

Check the four regions: $45 + 25 + 20 + 30 = 120$ — exercise only, both, diet only, neither.

Sets of Numbers

The Standard Number Systems

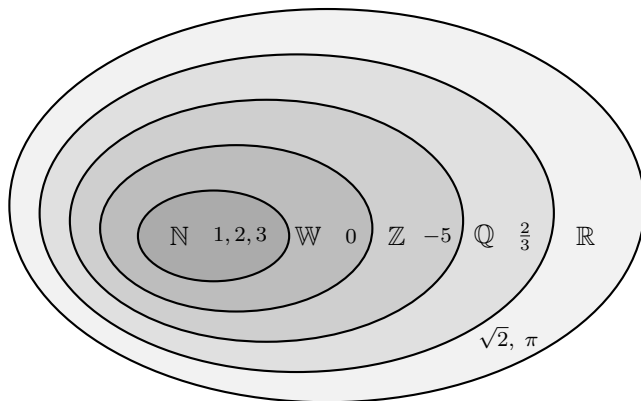
- **Natural numbers** $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ — the counting numbers.
- **Whole numbers** $\mathbb{W} = \{0, 1, 2, 3, 4, \dots\}$ — the naturals, plus zero.
- **Integers** $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ — the wholes, plus negatives.
- **Rational numbers** $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ — ratios of integers.
- **Real numbers** $\mathbb{R}$ — every rational **and** every irrational.

Each one sits inside the next

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

$\mathbb{Z}$ is from the German *Zahlen* (“numbers”); $\mathbb{Q}$ is for *quotient*.

The Number Systems, Nested



The outermost ring — inside $\mathbb{R}$ but outside $\mathbb{Q}$ — holds the **irrational** numbers: those that cannot be written as a ratio of integers, like $\sqrt{2}$ and π .

Rational vs. Irrational

A rational number's decimal expansion either stops or repeats

- $\frac{1}{2} = 0.5$ (terminates)
- $\frac{1}{3} = 0.3333 \dots$ (repeats)
- $\frac{1}{7} = 0.142857 142857 \dots$ (repeats, with period 6)

An irrational number's decimal expansion does neither

- $\sqrt{2} = 1.41421356 \dots$
- $\pi = 3.14159265 \dots$
- $e = 2.71828182 \dots$

A warning that Unit 1-4 will pay off

A computer stores only finitely many digits, so it never holds π , or even $\frac{1}{3}$, exactly. That is the subject of **floating-point arithmetic**.

Exercise

Try it

For each number below, name **every** one of $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ that it belongs to:

$$7, \quad -3, \quad 0, \quad \frac{5}{8}, \quad \sqrt{9}, \quad \sqrt{5}$$

Exercise: Solution

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	
7	✓	✓	✓	✓	✓	
-3			✓	✓	✓	negative
0		✓	✓	✓	✓	$\mathbb{N}$ starts at 1
$\frac{5}{8}$				✓	✓	a ratio of integers
$\sqrt{9}$	✓	✓	✓	✓	✓	$\sqrt{9} = 3$
$\sqrt{5}$					✓	irrational

The two traps

$\sqrt{9}$ is simply 3 — a square root is not automatically irrational. And 0 lands in $\mathbb{W}$ but not $\mathbb{N}$, since we take $\mathbb{N} = \{1, 2, 3, \dots\}$.

How Big Is an Infinite Set?

We defined $|A|$ as the number of elements in A . That works fine for $\{1, 2, 3\}$. But what is $|\mathbb{N}|$?

Counting runs out

You cannot count $\mathbb{N}$ element by element — you would never finish. Saying “the answer is infinity” does not help either, because we are about to find that **not all infinities are the same size**.

So we need a different idea of “same size”

One that never requires finishing a count — and that agrees with ordinary counting whenever the sets happen to be finite.

Same Size Means Perfect Pairing

How do you know a theater has as many seats as ticket-holders?

Everyone sits, every seat is filled, nobody doubles up — and you never counted either one.

Definition by pairing

Two sets have the **same cardinality** if their elements can be matched up one-to-one, with nothing left over on either side.

The unsettling example

Pair each natural number with its double:

$$1 \leftrightarrow 2, \quad 2 \leftrightarrow 4, \quad 3 \leftrightarrow 6, \quad 4 \leftrightarrow 8, \quad \dots \quad n \leftrightarrow 2n$$

Nothing is left over on either side, so $\mathbb{N}$ and the even numbers are **the same size** — even though the evens are a *proper subset* of $\mathbb{N}$. For infinite sets, a part can be as big as the whole.

Countable Infinity: $\aleph_0$

Countable

A set is **countable** if its elements can be arranged in an infinite **list** $a_1, a_2, a_3, \dots$ reaching every one — that is, paired off with $\mathbb{N}$.

This size is written $\aleph_0$ — “aleph-null,” the smallest infinity.

The integers are countable

Do not start at $-\infty$; zig-zag outward from 0:

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Every integer lands at a definite position, so $|\mathbb{Z}| = \aleph_0$.

So are the rationals

Arrange every fraction in a grid and sweep along diagonals, skipping repeats. So $|\mathbb{Q}| = \aleph_0$ — even though $\mathbb{Q}$ is dense on the line.

Cantor's Diagonal Argument

Can we list the **reals** the same way? Suppose we could — say, every real between 0 and 1:

$$r_1 = 0.\underline{5}23891 \dots$$

$$r_2 = 0.1\underline{4}7062 \dots$$

$$r_3 = 0.98\underline{5}134 \dots$$

$$r_4 = 0.310\underline{7}29 \dots$$

$$r_5 = 0.6274\underline{2}8 \dots$$

Build a number that cannot be there

Walk down the underlined diagonal and change every digit — if it is a 5 write 4, otherwise write 5:

$$d = 0.45455 \dots$$

d differs from r_n in the n th digit, so it matches **no** line of the list — yet it is a perfectly good real number between 0 and 1.

What the Diagonal Proves

The contradiction

We assumed the list contained **every** real between 0 and 1. The diagonal hands us one it missed. So no such list exists — and it is not a matter of trying harder, since the recipe defeats *any* list you propose.

The reals are uncountable

$\mathbb{R}$ cannot be paired off with $\mathbb{N}$. Its cardinality is strictly larger, and is written $\mathfrak{c}$ — “the continuum”:

$$\aleph_0 = |\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| < |\mathbb{R}| = \mathfrak{c}$$

Notice where the jump happens: adding negatives ($\mathbb{Z}$) and all the fractions ($\mathbb{Q}$) costs nothing. Adding the **irrationals** changes the size of infinity.

The Gap Between the Two

An honest open question

Is there a set strictly bigger than $\mathbb{N}$ but strictly smaller than $\mathbb{R}$ — some size wedged between $\aleph_0$ and $\mathfrak{c}$?

The guess that there is **no** such size is the **continuum hypothesis**.

The answer is stranger than yes or no

Gödel and Cohen proved that the continuum hypothesis can be neither **proved** nor **disproved** from the standard axioms of set theory. Both answers are consistent. You may assume either one.

A reminder that “how many?” is a deeper question than it looks — and that the number line you have used since grade school still holds open problems.

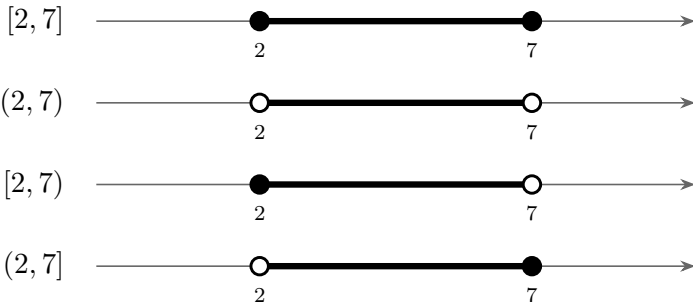
Intervals

Intervals, and How We Write Them

Interval

An **interval** is the set of **all** real numbers lying between two given numbers, called the **endpoints**.

The only decision is whether the endpoints themselves are **included**. A **square** bracket includes; a **round** bracket excludes. Here are all four combinations on 2 and 7:



Every Interval, Three Ways

Interval	Number line	Set-builder
(a, b)		$\{x \mid a < x < b\}$
$(a, b]$		$\{x \mid a < x \leq b\}$
$[a, b)$		$\{x \mid a \leq x < b\}$
$[a, b]$		$\{x \mid a \leq x \leq b\}$

The bracket, the dot, and the inequality sign all carry *exactly* the same information.

Intervals That Run Forever

∞ is **not a number** — it is shorthand for “without end.” Its side therefore **always** gets a round bracket: there is no endpoint there to include.

Interval	Number line	Set-builder
(a, ∞)		$\{x \mid x > a\}$
$[a, \infty)$		$\{x \mid x \geq a\}$
$(-\infty, b)$		$\{x \mid x < b\}$
$(-\infty, b]$		$\{x \mid x \leq b\}$
$(-\infty, \infty)$		$\mathbb{R}$

Reading Intervals in Practice

Closed — a lab reference range

“4.0 to 5.6, inclusive” is $[4.0, 5.6]$. A patient at exactly 5.6 is **in** range.

Half-open — an age bracket

“40 to 49” really means $[40, 50)$ — you leave the bracket the moment you turn 50. Half-open intervals tile the number line with no gaps and no overlaps, which is why nearly every bracketed category works this way.

Open — a strict threshold

“Anything above 100, but not 100 itself” is $(100, \infty)$.

Watch the wording: *at least* and *no more than* include the endpoint; *greater than* and *under* exclude it. “Between” says nothing either way — so it always needs pinning down.

Three Ways to Say the Same Thing

Being able to move between these three fluently is the whole skill.

Inequality	Set-builder	Interval
$2 < x < 7$	$\{x \in \mathbb{R} \mid 2 < x < 7\}$	$(2, 7)$
$2 \leq x \leq 7$	$\{x \in \mathbb{R} \mid 2 \leq x \leq 7\}$	$[2, 7]$
$x \geq 2$	$\{x \in \mathbb{R} \mid x \geq 2\}$	$[2, \infty)$
$x < 7$	$\{x \in \mathbb{R} \mid x < 7\}$	$(-\infty, 7)$

Exercise

Try it

A test result is in range when it is **at least** 70 and **less than** 100.

Write that set all three ways — as an inequality, in set-builder notation, and as an interval — then draw it on a number line.

Exercise: Solution

“At least 70” includes 70, so that end is **closed**. “Less than 100” excludes 100, so that end is **open**.

Inequality	$70 \leq x < 100$
Set-builder	$\{ x \in \mathbb{R} \mid 70 \leq x < 100 \}$
Interval	$[70, 100)$

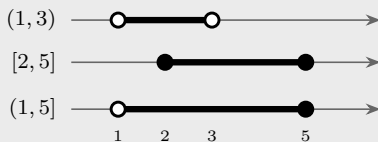


A result of exactly 70 is in range; a result of exactly 100 is not.

Operations on Intervals

The set operations apply to intervals too — draw both, then read off the answer.

Union: $(1, 3) \cup [2, 5] = (1, 5]$



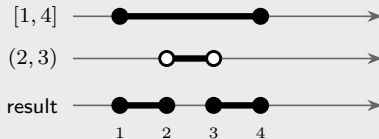
They overlap, so they merge into one stretch: open at 1, closed at 5.

Intersection: $(1, 3) \cap [2, 5] = [2, 3)$

Only the overlap survives. 2 lies in both, so it is included; 3 is in neither, so it is not.

Difference of Intervals

Difference: $[1, 4] \setminus (2, 3) = [1, 2] \cup [3, 4]$

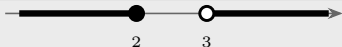


Removing an *open* middle leaves two *closed* pieces — 2 and 3 were never in $(2, 3)$, so they were never removed.

Watch the endpoints. That is where every mistake happens.

Unbounded Examples

A union that does *not* merge: $(-\infty, 2] \cup (3, \infty)$



No overlap, so nothing merges — the union is genuinely two separate pieces, and this is already its simplest form. An “abnormal lab value” set looks exactly like this: too low, or too high.

Intersection: $(-\infty, 4) \cap (2, \infty) = (2, 4)$

Numbers that are both less than 4 and greater than 2. Neither endpoint belongs to both, so both stay open.

Complement within $\mathbb{R}$: $\mathbb{R} \setminus [1, 3] = (-\infty, 1) \cup (3, \infty)$

Removing a closed interval leaves open ends: 1 and 3 were inside, so they are now excluded.

Quantifiers

Saying “All” and “Some”

We have already met a statement whose truth depends on a variable: “ $x \in A$ ” is true for some x and false for others. Plain numbers behave the same way — “ $x \geq 0$ ” is true when $x = 3$, false when $x = -7$.

To state a general rule about such a statement, name it and **tag it with the variable inside**:

$$p(x) = “x \geq 0”$$

The parentheses are pure bookkeeping — they only record that an x sits inside.

On its own $p(x)$ has no truth value. Two ways to give it one:

- Claim it holds for **every** x . (universal)
- Claim it holds for **at least one** x . (existential)

The two quantifiers

Symbol	Name	Read as
$\forall$	universal quantifier	“for all” / “for every”
$\exists$	existential quantifier	“there exists” / “for some”
$\nexists$	—	“there does not exist”

The Universal Quantifier $\forall$

$\forall$ states that something is true of **every** element of a set.

Example

$$\forall x \in \mathbb{R}, x^2 \geq 0$$

Read: “for every real number x , x squared is greater than or equal to 0.”
(True — squaring never gives a negative.)

More examples

- $\forall x \in \mathbb{N}, x \geq 1$ (true — the naturals start at 1)
- $\forall x \in \mathbb{R}, x^2 > 0$ (false — $x = 0$ ruins it)
- $\forall x \in \mathbb{Z}, x \in \mathbb{Q}$ (true — every integer n is the fraction $n/1$)

One counterexample is enough to make a $\forall$ statement false.

The Existential Quantifier $\exists$

$\exists$ states that there is **at least one** element for which something is true.

Example

$$\exists x \in \mathbb{R} \text{ such that } x^2 = 2$$

Read: “there exists a real number x whose square is 2.” True — and notice that $x = \sqrt{2}$ is irrational, so the same claim over $\mathbb{Q}$ would be **false**.

To say that no such element exists, strike through the symbol: $\nexists$.

Example

$$\nexists x \in \mathbb{R} \text{ such that } x^2 = -1$$

Read: “there does not exist a real number x whose square is -1 .”

One example is enough to make an $\exists$ statement true.

The Set Matters

A quantified statement is only as good as the set it ranges over. The **same statement** can flip its truth value when the set changes:

Claim	Truth value
$\exists x \in \mathbb{R}$ such that $x^2 = 2$	TRUE ($x = \sqrt{2}$)
$\exists x \in \mathbb{Q}$ such that $x^2 = 2$	FALSE ($\sqrt{2}$ is irrational)
$\forall x \in \mathbb{N}, x > 0$	TRUE
$\forall x \in \mathbb{Z}, x > 0$	FALSE ($x = -5$)

Always name the set

“ $\forall x$ ” with no set attached is incomplete. Writing $\forall x \in A$ says exactly what you are quantifying over — which is why sets had to come first.

Exercise: Translating Quantifiers

Part 1 — say each of these out loud, in plain English

- ① $\forall x \in \mathbb{Z}, x^2 \geq 0$
- ② $\exists x \in \mathbb{N}$ such that $x > 100$
- ③ $\nexists x \in \mathbb{R}$ such that $x^2 < 0$

Part 2 — now go the other way, into symbols

- ① Every rational number is also a real number.
- ② Some integer is negative.
- ③ There is no natural number less than 1.

Exercise: Solution

Part 1 — into English

- ① “for every integer x , x squared is at least 0.” (true)
- ② “there is a natural number greater than 100.” (true — 101)
- ③ “no real number has a negative square.” (true)

Part 2 — into symbols

- ① $\forall x \in \mathbb{Q}, x \in \mathbb{R}$
- ② $\exists x \in \mathbb{Z}$ such that $x < 0$
- ③ $\nexists x \in \mathbb{N}$ such that $x < 1$

The last one can also be written $\forall x \in \mathbb{N}, x \geq 1$ — “none are below 1” and “all are at least 1” say the same thing. That is the next slide.

Negating Quantifiers

Negation swaps the two quantifiers — the same flip we saw in De Morgan's laws in Unit 1-1.

Negation rules

$$\neg(\forall x, p(x)) = \exists x \text{ such that } \neg p(x)$$

$$\neg(\exists x \text{ such that } p(x)) = \forall x, \neg p(x)$$

Why

- If $p(x)$ is **not** true for every x , then **some** x must break it.
- If **no** x makes $p(x)$ true, then **every** x makes it false.

This is De Morgan again: $\forall$ is a long AND, $\exists$ is a long OR, and negation flips one into the other.

Negation Examples

Negating a $\forall$

$$\begin{aligned}\neg(\forall x \in \mathbb{R}, x^2 > 0) \\ = \exists x \in \mathbb{R} \text{ such that } x^2 \leq 0\end{aligned}$$

(And indeed — take $x = 0$.)

Negating an $\exists$

$$\begin{aligned}\neg(\exists x \in \mathbb{Q} \text{ such that } x^2 = 2) \\ = \forall x \in \mathbb{Q}, x^2 \neq 2\end{aligned}$$

The **inner statement** is negated too: $>$ becomes $\leq$, and $=$ becomes $\neq$.

Exercise

Try it

Negate the claim

"Every patient in the cohort had a follow-up visit."

Then answer: what exactly would you have to find in order to **disprove** it?

Exercise: Solution

Let C be the cohort and $f(x) = "x \text{ had a follow-up visit}":$

$$\forall x \in C, f(x)$$

Now negate, using $\neg(\forall x, p(x)) = \exists x \text{ such that } \neg p(x):$

$$\neg(\forall x \in C, f(x)) = \exists x \in C \text{ such that } \neg f(x)$$

In words

*"There is at least one patient in the cohort who did **not** have a follow-up visit."*

What it takes to disprove the original

Exactly **one** such patient — a single **counterexample**. You do not have to say anything about the rest of the cohort, and finding a hundred patients who *did* follow up proves nothing either way.

Back to the Cohort

Unit 1-1 opened with a criterion we could write in logic but not yet *collect*:

The English

"Patients who are diabetic AND (hypertensive OR obese), but NOT pregnant."

Let U be all patients in the study, and name each property as a statement about x :

$d(x)$ = " x is diabetic"

$h(x)$ = " x is hypertensive"

$o(x)$ = " x is obese"

$g(x)$ = " x is pregnant"

The cohort is a set

$$\text{Cohort} = \{ x \in U \mid d(x) \wedge (h(x) \vee o(x)) \wedge \neg g(x) \}$$

Logic supplies the rule; set-builder notation turns it into a **collection you can count**.

Ordered Pairs and Tuples

When Order Matters

In a set, order is irrelevant: $\{2, 5\} = \{5, 2\}$. But for a point in the plane, order is everything — $(2, 5)$ and $(5, 2)$ are different places.

Ordered pair

An **ordered pair** (a, b) is a collection of two elements in which **the order matters**. Two ordered pairs are equal exactly when

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

Notation warning

$(2, 5)$ is also how we write an open interval. **Context tells you which is meant** — a pair of coordinates, or a stretch of the number line. Watch for it.

Tuples

n -tuple

An **n -tuple** $(a_1, a_2, \dots, a_n)$ generalizes the ordered pair to any finite number of entries, order still mattering.

- A **2-tuple** is an ordered pair: (a, b) .
- A **3-tuple** (triple): (a, b, c) .
- A **4-tuple**: (a, b, c, d) . And so on.

Two n -tuples are equal exactly when they agree entry by entry: $a_i = b_i$ for every i .

Examples

- $(2, 5) \neq (5, 2)$ — order matters.
- $(1, 2, 3) \neq (3, 2, 1)$ — same reason.
- $(x, y, z) = (4, -2, 7)$ locates a point in three-dimensional space.

Why Tuples Matter Here

They are the bridge to the rest of the course

- **Coordinates:** every point on a graph is a tuple.
- **Relations and functions (Unit 1-3):** a relation is literally a *set of ordered pairs* (*input, output*).
- **Vectors (Unit 2):** a vector in $\mathbb{R}^n$ is an n -tuple of numbers.
- **Data:** one row of a data table — one patient's measurements — is a tuple. A whole dataset is a set of tuples.

The takeaway

A **set** answers “which things?” A **tuple** answers “which things, in which slots?” Nearly all of data science is built out of the two.

Review

Review

What we covered

- ① A **set** is a collection of elements; $\{x \mid \dots\}$ describes one by a rule.
- ② $A \subseteq B$ means every element of A is in B ; $\emptyset$ is a subset of everything.
- ③ $A \cup B$, $A \cap B$, $A \setminus B$, $\bar{A}$ are the set versions of OR, AND, AND NOT, NOT — **De Morgan** included.
- ④ **Venn diagrams** organize the regions; $|A \cup B| = |A| + |B| - |A \cap B|$ counts them.
- ⑤ $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; rational decimals terminate or repeat, irrational ones do not.
- ⑥ $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all **countable** ($\aleph_0$); Cantor's diagonal shows $\mathbb{R}$ is strictly bigger.
- ⑦ **Intervals** are stretches of the real line; brackets record the endpoints.
- ⑧ **Quantifiers** $\forall, \exists$ turn a statement about x into a claim about a whole set; negation swaps them.
- ⑨ **Tuples** are collections where order matters — points, data rows, vectors.